# NEW WAYS TO PLAY: A STUDY IN ALTERNATIVE GAME CONTROLLERS

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## Introduction & Background

Picked this project because of the knowledge in multiple controls like VR, Wii Remote

Explores alternative game controllers.

How it helps shape player immersion, experience and accessibility

Motion-Based Controller for the study

## Problem Statement

Problem Statement:

- Expensive Controllers
- Not all Controllers Cater to everyone.
- Limited accessibility creates barriers for many players.
- Environmental Impacts

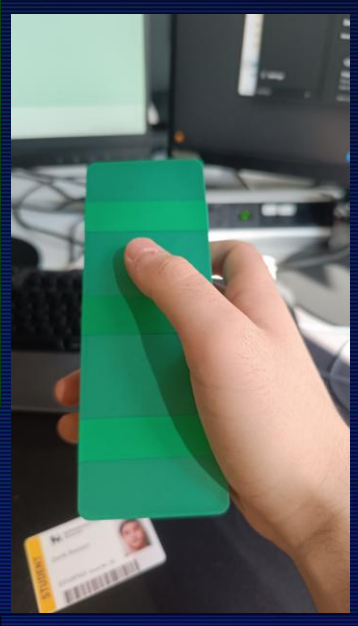
HP

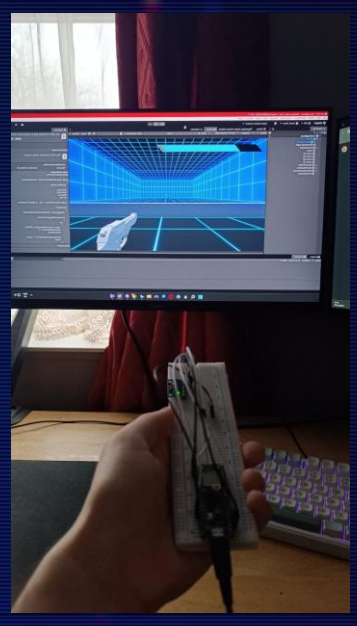
## Aims & Objectives

- Develop a motion-based controller and game to evaluate alternative controllers.
- Create a Low-cost Controller
- Analyses usability, responsiveness, immersion and accessibility
- Does a Motion-Based Controller Improve experience

## Methods

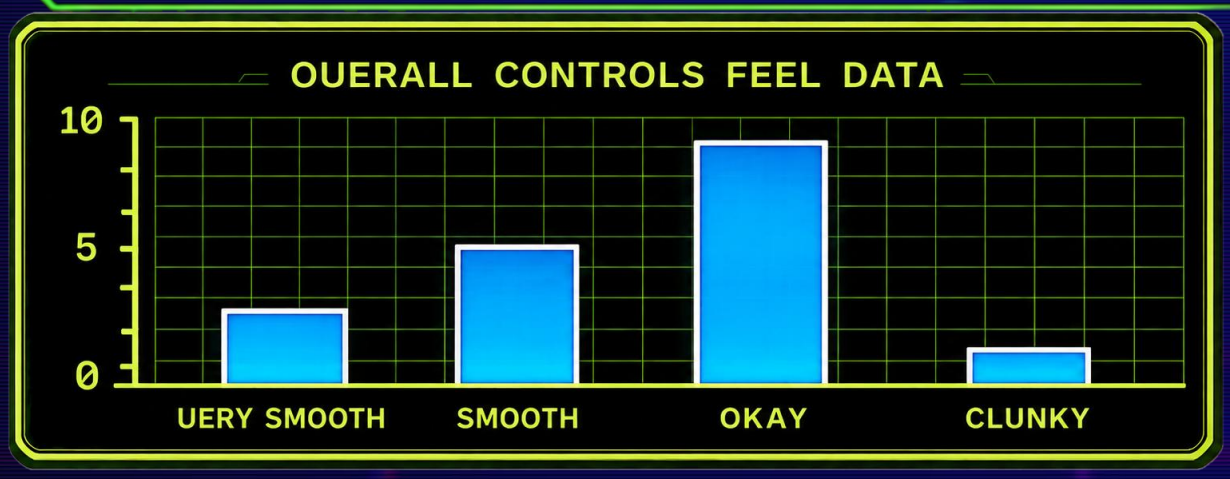
- Raspberry Pi Pico WH
- MPU6050 (gyro/accelerometer)
- Breadboard, jumperwires, buttons and 3D printed Case
- Unity + C# + Thonny
- Data Gathering (13 Testers)





## Results / Inplementation & Testing / Evaluation

- Data was gathered in person there were 13 participants to conduct the test with consent
- Play the game using the motion-based controller
- Participants found the controller to be fun but could use with further improvements to the 3D printed case design and gyro
- By enabling motion controls, players can control the on-screen cursor to look around the game as intended but alsos they can navigate the UI and interface of any application and OS, extending functionality beyond just the game environment.




## Discussion

- Purchased the required componenbts after in-depth research
- Using Thonny and CircruitPython to have the components fully functional and working together
- Created a prototype and also making the final version
- Developed a Unity game within Unity using C# to help showcase the motion-based controller
- And finally conducted tests for data on the controller

## Conclusion and Future work

In conclusion the project itself was succesful on gathering data on alternative game controllers and meetings its core objectives. As well as inportant information that could be used for future work, research and development. Furthermore this product can be inproved much more and fixing the key issues that were addressed during the tests.

## References

Malone, L. A. et al., 2023. Usability of the GAIMPLANK video game controller for people with mobility impairments: Observational study. JMIR Serious Games, Volume 11, p. e38484.

Skalski, P. et al., 2010. Mapping the road to fun: Natural video game controllers, presence, and game enjoyment. New Media & Society, 12 October, 13(2), pp. 224-242.

Thanks to all the support from Richard Davies during this project |
https://github.com/MamudRostam/Mamud_Final_Year_Project.git | QR Code